Final Project Memo – Software and Methods

For the final project, I created an audio-visual experience with many VSTi reaper instruments. The specialized instruments were coded in a language called PureData (PD), which allows signal processing modules to be routed together to process audio streams. The instrument patches are modified to allow MIDI inputs, converting the notes from the channel MIDI stream into a MIDI note message, which is then processed into a frequency in PD through "mtof." The basic idea then is to fully utilize the stereo channels of the piece to draw shapes on an oscilloscope. In the attached PowerPoint presentation, I used an oscilloscope I borrowed from the parts room to play the audio on the x and y channels to project the output of the audio chain. While prototyping, however, I did not always connect the oscilloscope to my laptop. I instead opted to use an open-source software called "Oscilloscope!" to monitor my audio output and view the shapes it produced. From that point on, it took a long process of gently tweaking parameters in Reaper automation and playing them back while monitoring the oscilloscope as the shapes changed. In Figure 2, the exploded Reaper view shows this process.

I started the composition with dots contracting and expanding into a circle, and I did this by varying the size and number of dashes in the plugin shown in Figure 1. For the second portion I used two oscillators with an integer multiple to create these Lissajous curves, then I used the "JS: Ring Modulator" to add a "third dimension" to the figures. I also added an ADSR envelope from the PD-ELSE library The multiple was parameterized in Reaper, as well as the MIDI notes for all tracks and plugins.

I then generated cubes in a similar fashion, varying their size and parameters with custom automation items. For the next segment I used the mushroom & butterfly plugin from oscilloscopemusic.com and again varied the parameters allowing for rotation and the growth of more mushrooms. Then I played with the volume and other parameters to shape and change them. The one thing I dislike about this plugin is how shrill the finished product is, and that is a result of the sharp edges in the waveform and lots of high-frequency content.

Finally, I added an array of circles interspersed with the mushrooms, which led to the finale of the letters: e-music! I used a fun website to generate these waveforms, it allows you to make a sketch and generate a tone depicting this 2D object: https://bummsn.de/osc_txt/. I only later found this cool plugin that allowed me to generate json coordinates from svg files: https://spotify.github.io/coordinator/

The plugins I used were a part of a tutorial series on Oscilloscope Music, and made available through GitHub at: https://github.com/Eric-Lennartson/oscilloscope-music-tutorials. The additional patch was the “mushrooms & butterflies” patch from: https://oscilloscopemusic.com/software/puredata/. To integrate the PureData language with Reaper, I used the compatibility VSTi/standalone app, found at https://plugdata.org. Additionally, I was inspired by https://www.youtube.com/@ChrisAllenMusic and https://www.youtube.com/@ericlennartson/ with his guide to Oscilloscope Music and aforementioned documentation. The PD-Else library was pre-installed with PlugData, but the documentation can be found at: https://github.com/porres/pd-else.

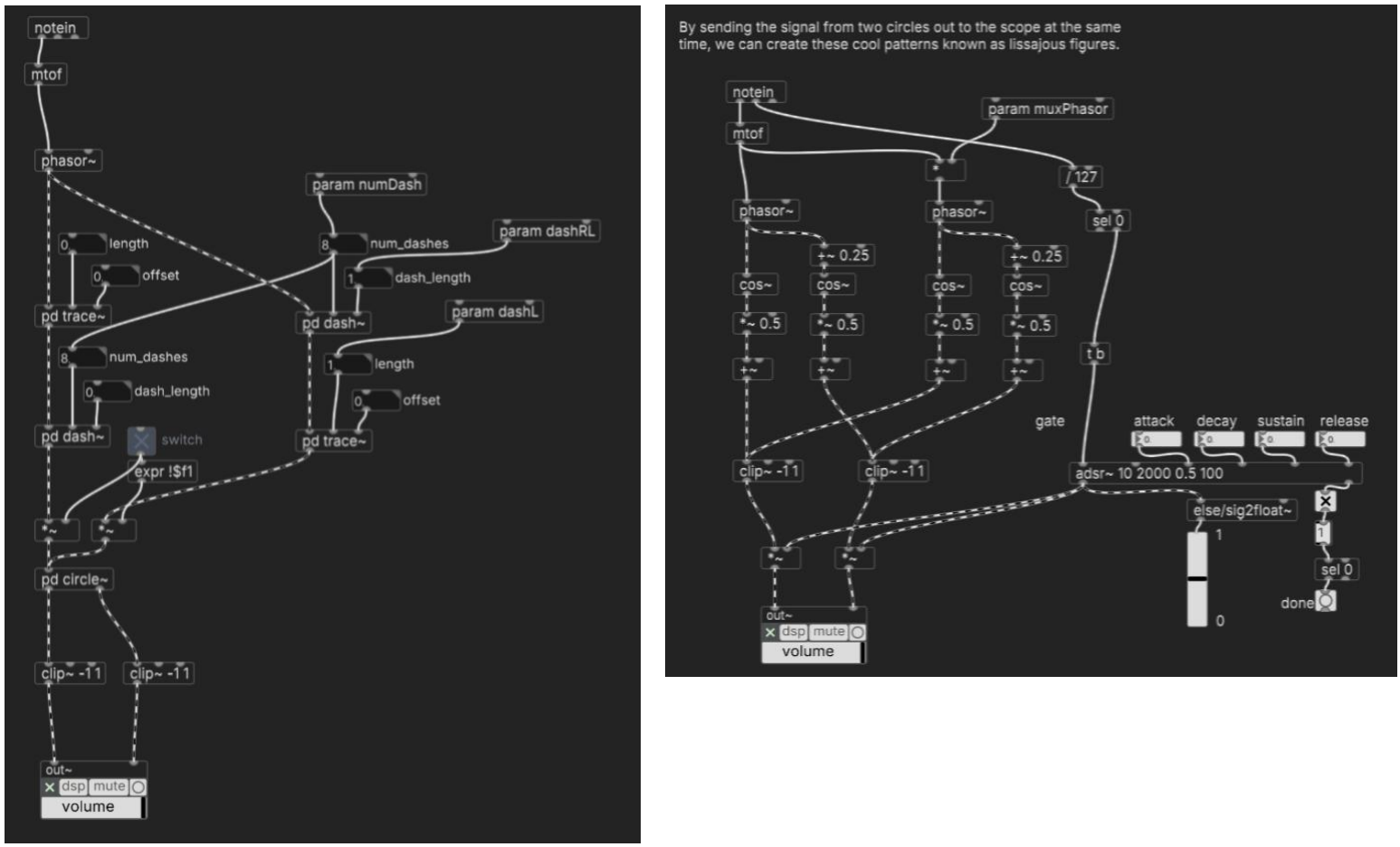

Figure 1 Dash and Trace PD File (Left) and Lissajous Figure Generator PD (Right)

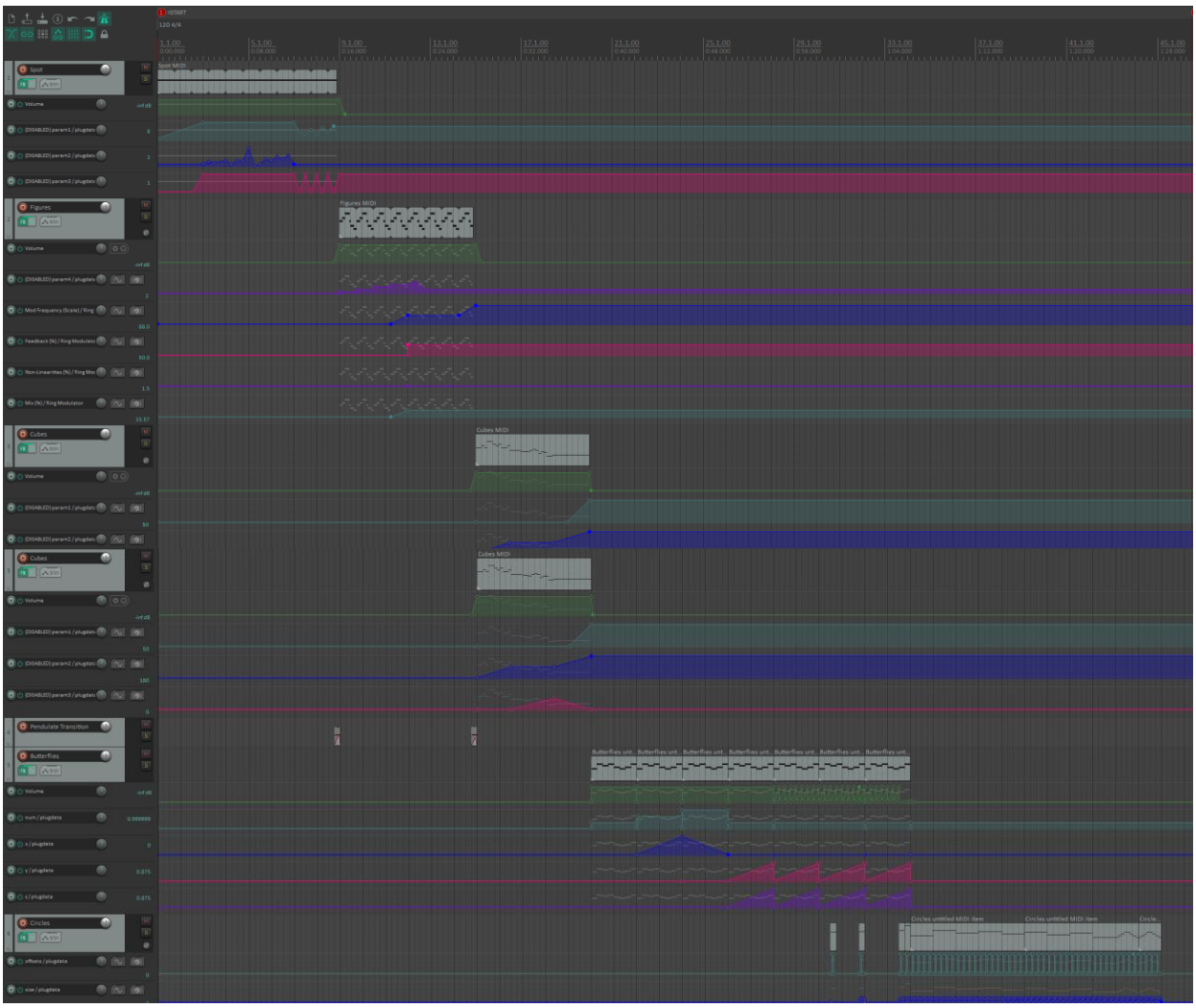

Figure 2 Reaper Automation

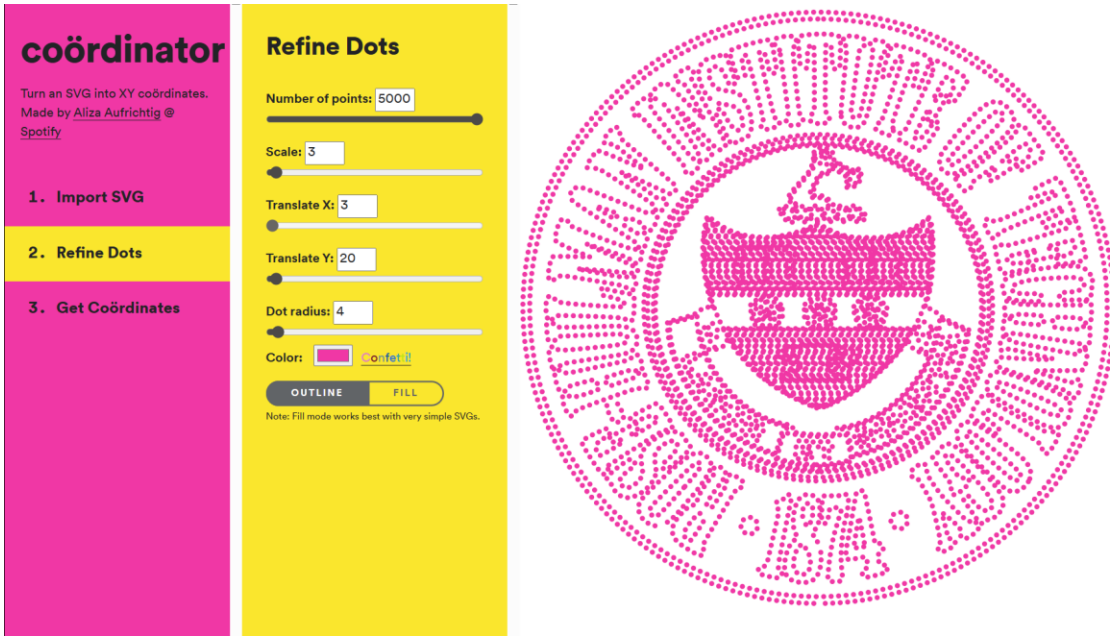

Figure 3 SVG to JSON Tool

Figure 4 Output JSON to Oscilloscope Signals

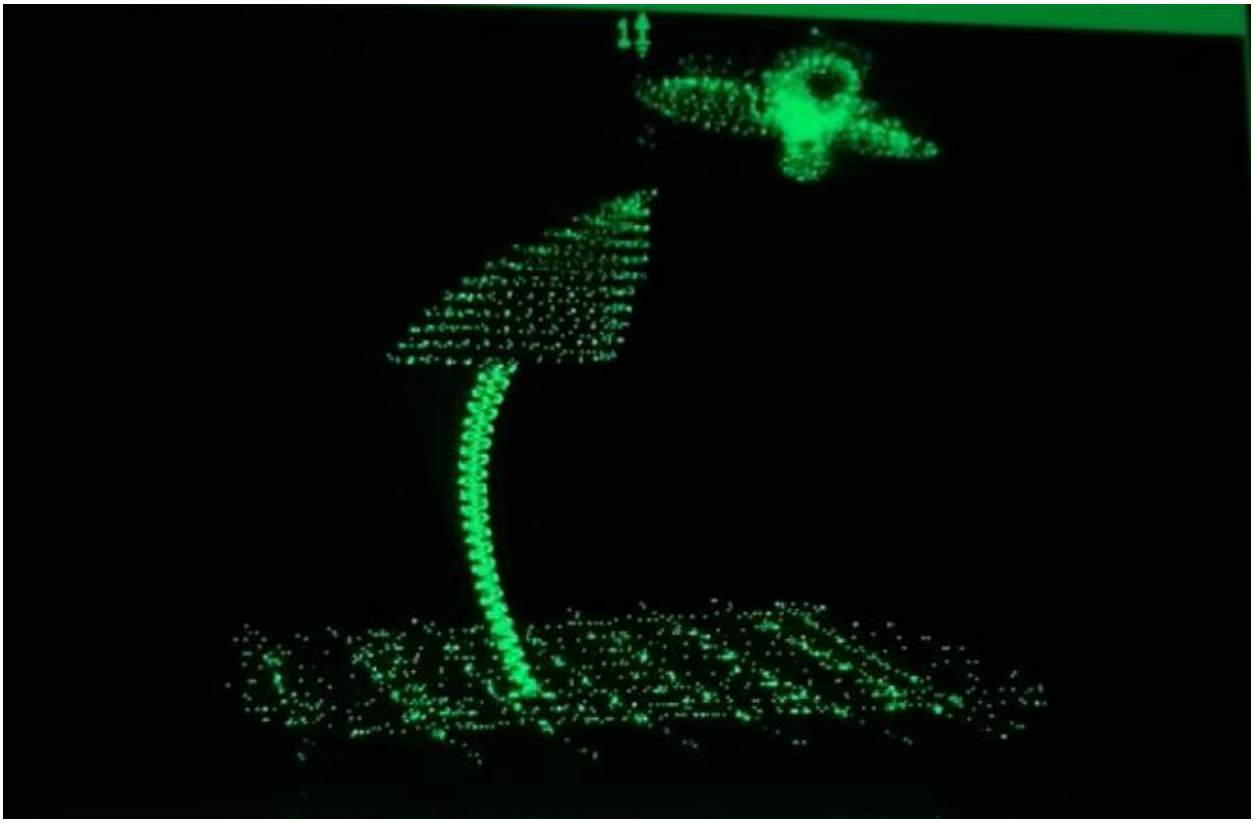

Figure 5 Resulting Mushroom and Butterfly Oscilloscope Patch